

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

NUMERICAL METHODS FOR ENGINEERS

(Common to CE, EEE, ME, ECE, CSE, IT, FT, AI&DS, CSE (AI), CSE (DS), CSE (AI&ML), CSE (IoT), CSE (CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- Using the Bisection Method, find the first approximation to the root of the equation $f(x) = x^3 - 4x + 1$ in the interval [1, 2]. 2M
 - What is the key condition for the convergence of the Gauss-Seidel method? 2M
 - State the principle of least squares and explain its importance in curve fitting. 2M
 - Given the data points (1, 2), (2, 3) and (3, 5) determine the equation of the best-fit linear curve using the method of least squares. 2M
 - In Newton's Forward Interpolation Formula, what type of differences are used. 2M
 - Given the data points $x_2 = 3$, $y_2 = 6$, $x_3 = 4$, $y_3 = 8$, find the backward difference ∇y_3 . 2M
 - State the main difference in application between Simpson's 1/3 Rule and Simpson's 3/8 Rule. 2M
 - Using Simpson's 1/3 Rule, estimate the integral of $f(x)$ from $x = 0$ to $x = 2$ with the function values $f(0) = 1$, $f(1) = 4$, and $f(2) = 9$. 2M
 - What is the general formula for the fourth-order Runge-Kutta method to solve $dy/dx = f(x, y)$. 2M
 - For the differential equation $dy/dx = y+x$ with $y(0) = 1$, apply one iteration of the Modified Euler's Method to estimate $y(0.1)$ with a step size of $h = 0.1$. 2M

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 A real root of the equation $f(x) = x^3 - 5x + 1 = 0$ lies in the interval (0, 1). Perform four iterations of the Regula Falsi method to obtain this root. 10M

OR

- 3 Solve by Gauss Seidel method 10M
- $$27x + 6y - z = 80$$
- $$6x + 15y + 2z = 70$$
- $$x + y + 50z = 100.$$

- 4 Fit a Straight line $y = ax + b$ in the least square sense for the data : 10M

x	5	10	15	20	25
y	16	19	23	26	30

OR

- 5 Fit a curve of the form $y = ab^x$ in the least square sense for the following data: 10M

x	0	2	4	5	7	10
y	100	120	256	390	710	1600

Contd. in Page 2

- 6 The areas y of circles for different diameters x are given below;

10M

x	80	85	90	95	100
y	5026	5674	6362	7088	7854

Calculate the area when $x = 98$.

OR

- 7 The following table gives the viscosity of an oil as a function of temperature. Use Lagrange's formula to find viscosity of oil at a temperature of 140° .

10M

temp°	110	130	160	190
viscosity	10.8	8.1	5.5	4.8

- 8 Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's 1/3 rd rule taking four equal strips and hence deduce an approximate value of π .

10M

OR

- 9 Evaluate $\int_0^1 \frac{dx}{1+x}$ taking seven ordinates by applying Simpson's 3/8th rule. Hence deduce the value of $\log_e 2$.

10M

- 10 Using modified Euler's method find y at $x = 0.2$ given $\frac{dy}{dx} = 3x + \frac{y}{2}$ with $y(0) = 1$ taking $h = 0.2$.

10M

OR

- 11 Use fourth order Runge-Kutta method to find y at $x = 0.1$ given that $\frac{dy}{dx} = 3e^x + 2y$, $y(0) = 0$ & $h = 0.1$.

10M

B.Tech IV Year I Semester (R19) Supplementary Examinations December 2024

NUMERICAL METHODS FOR ENGINEERS

(Common to all branches)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- (a) Explain the Bisection Method of finding first approximate real root of $x^3 + x - 1 = 0$. 2M
- (b) Find first approximate root of the equation $x^3 - 3x + 4 = 0$ using Regula falsi method, which lie between -3 and -2. 2M
- (c) Write the normal equations to fit the parabola of the form $y = a + bx + cx^2$. 2M
- (d) Write the normal equations to fit the equation of the form $y = ae^{bx}$. 2M
- (e) State the importance of Bessel's formula. 2M
- (f) Explain when the Gauss forward and backward interpolation formulae are applied to the given data. 2M
- (g) Explain the working rule to evaluate the given definite integrals by Trapezoidal rule. 2M
- (h) What is the difference between Simpson's one third rule and Simpson's three eighth rule? 2M
- (i) Justify your answer that "Taylor's series is applicable for non differentiable functions". 2M
- (j) What do you understand by initial value problem? 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Compute the real root of the equation $\cos x = 3x - 1$, by the method of Regula falsi method, correct to three decimal places, carryout three approximations. 4M
- (b) Find the real root of the equation $3x = \cos x + 1$ correct to three decimal places using Newton's Raphson method. 6M

OR

- 3 (a) Using Regula-falsi method compute the real root of the equation $xe^x = \cos x$, correct to three decimal places. 4M
- (b) Apply Gauss Jordan method to solve the system of equations: 6M
 $x + y + z = 0$, $x + 2y + 3z = 2$, and $2x - y + 3z = 10$.

- 4 (a) Fit a straight line to the following data: 5M

x	1961	1971	1981	1991	2001
y	8	10	12	10	16

- (b) Fit a parabola $y = a + bx + cx^2$ to the following data: 5M

x	2	4	6	8	10
y	3.07	12.85	31.47	57.38	91.29

OR

Contd. in Page 2

- 5 (a) A simply supported beam carries a concentrated load P (lb) at its mid point. Corresponding to various values of P , the maximum deflection Y (in) is measured. The data are given below: 4M

P	100	120	140	160	180	200
Y	0.45	0.55	0.6	0.70	0.80	0.85

Find the law of the form $Y = a + bP$.

- (b) Find the curve of the form $y = ae^{bx}$ that fits the best the following data: 6M

x	3	4	5	6	7
y	6	9	10	11	12

- 6 (a) Using Newton's appropriate interpolation formula, find the values of y at $x = 8$ and at $x = 22$ from the following table: 4M

x	0	5	10	15	20	25
y	7	11	14	18	24	32

- (b) Using Lagrange's interpolation formula, fit a polynomial which passes through the points $(-1, 0), (1, 2), (2, 9), (3, 8)$. 6M

OR

- 7 (a) Interpolate by means of Gauss interpolation formula the population of a town for the year 1974, given that: 5M

Year	1939	1949	1959	1969	1979
Population in thousands	9	10	11	13	15

- (b) Find the interpolation polynomial using Lagrange's interpolation formula to the data given below and hence find $f(3)$. 5M

x	0	1	2	5
y	2	3	12	147

- 8 (a) Evaluate $\int_2^7 \frac{1}{x} dx$ using Trapezoidal rule by taking 5 equal sub intervals. 4M

- (b) Evaluate $\int_0^{0.6} e^{-x^2} dx$ by taking seven ordinates and by Simpson's 1/3 rule. 6M

OR

- 9 (a) Evaluate $\int_{0.2}^{1.4} (\sin x - \log x + e^x) dx$ by taking seven ordinates using Simpson's 1/3 rule. 5M

- (b) Evaluate $\int_4^{5.2} \log_e x dx$ by taking six equal strips and by Simpson's 3/8th rule. 5M

- 10 (a) Employ Taylor's series method to obtain approx. value of y at $x = 0.2$ for the differential equation $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$. 4M

- (b) Using Modified Euler's method, Find an approximate value of y when $x = 0.2$ given that $\frac{dy}{dx} = x + y$ and $y = 1$ when $x = 0$. 6M

OR

- 11 (a) Solve $y' = x + y$ given $y(1) = 0$. Find $y(1.1)$ by Picard's method. 5M

- (b) Apply Runge - Kutta method of fourth order to find an approximate value of y when $x = 0.2$ given that $\frac{dy}{dx} = x + y$ and $y = 1$ when $x = 0$. 5M
